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A LOWER ORDOVICIAN PALEOMAGNETIC POLE FROM THE ONEOTA DOLOMITE, UPPER MISSISSIPPI RIVER VALLEY

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Abstract. Of 149 oriented specimens of the lower Ordovician Oneota dolomite subjected to stepwise thermal or alternating-field demagnetization, 50 contained a characteristic shallow east-southeasterly or west-northwesterly component of magnetization identifiable on orthogonal vector projections. For 48 other specimens the demagnetization trajectories defined great-circle paths which intersect close to the characteristic direction. The site-mean directions for the characteristic component were determined by using both the characteristic directions and great-circle data, and these yield a formation-mean direction obtained from 10 sites with a declination of 105°, an inclination of +2°, and a precision parameter k of 17.6 (α95 = 11.9°). The mean of the site virtual poles falls at 10.4° N, 166.4° E (K = 48.0, A95 = 7.0°), which is close to previously published late Cambrian paleopoles. Both polarity groups have predominantly positive inclinations, suggesting the presence of an uncleaned downward component whose unblocking-temperature and coercivity spectra overlap very strongly with those of the characteristic magnetization. We interpret this downward component as a drilling-induced remanent magnetization, and we hope to have removed its contribution by averaging the dual-polarity results. Multicomponent magnetizations were also present in samples collected from blocks of Oneota dolomite incorporated into the overlying New Richmond sandstone (lower Ordovician), but only six out of eight samples yielded well-defined great-circle paths. The best fit intersection of these defines a northerly and steep recent component, while the directions of the higher-stability components are different for each block. Due to the small number of samples and imperfect resolution of magnetic directions, the conglomerate test is not conclusive; nevertheless it suggests that the stable remanence in the Oneota dolomite was acquired prior to the deposition of the New Richmond sandstone.

Introduction

While the broad outline of lower Paleozoic apparent polar wander (APW) with respect to the North American craton has been known for at least ten years [e.g., McElhinny, 1973], substantial differences of opinion still exist concerning the spatial details and time calibration of this part of the APW path. Published paleomagnetic poles for the Cambrian and Ordovician are distributed along an arc roughly 90 degrees away from the craton, extending northwestward from a point near the equator at 180° E, to a well-defined cluster of poles near 35° N, 120° E [Van Alstine, 1979; Irving, 1979; Watts et al., 1980; Van der Voo, 1981; Gillett, 1982]. This distribution indicates equatorial paleolatitudes for North America for most of the early Paleozoic, but leaves open the question of orientation with respect to the paleomeridian. The fact that the northwestern end of the arc approaches the late Paleozoic section of the APW path makes it difficult to rule out displacement of some poles in this direction by subsequent remagnetization. Furthermore, distribution of paleopoles in an arc centered on the sampling locality suggests that tectonic rotations or complex deformational histories [MacDonald, 1980] may be important in some of these results. There is thus a very large degree of uncertainty in the temporal progression of APW along the streaked distribution of paleopoles, and conflicting interpretations have been offered. A loop in the APW path during the Cambrian has been proposed, spanning the distribution of poles from southeast to northwest and back [Watts et al., 1980]. The validity of the northwestern Middle Cambrian poles has been subject to question, as conclusive evidence against remagnetization has not yet been obtained [Lynnes and Van der Voo, 1984]. Gillett [1982] has reviewed published late Cambrian paleomagnetic results from cratonic sediments and concluded that remagnetization and/or rotation could not be conclusively ruled out in any of them. A number of paleopoles from Upper Cambrian-lower Ordovician mafic and alkalic igneous rocks of Colorado fall near the northwestern end of the streak [Larson and Mutschler, 1971; Patterson et al., 1978; Larson et al., 1984]. Larson et al. [1984] therefore questioned the validity of the southeastern late Cambrian poles, proposing that the location of the Cambro-Ordovician boundary along the APW path is near 35° N, 120° E, and that APW during the Ordovician was very minor. This interpretation is also subject to the criticism that no conclusive evidence against remagnetization was presented.

Some of the room for disagreement in this matter is provided by the lack of reliable results from lower Ordovician rocks. Our objective in this investigation has been to determine a well-defined early Ordovician paleomagnetic pole for North America, with a minimum of uncertainty due to possible remagnetization or tectonic rotation.

The lower Ordovician stratigraphy of North America is dominated by extensive carbonate platforms, a fact which supports the equatorial paleolatitudes inferred from lower Paleozoic paleomagnetic data [Scotese, 1984]. However, paleomagnetic analysis of carbonates is difficult for a number of reasons, foremost among which is the very low intensity of natural remanent magnetization (NRM) typical of carbonates. Further, recent work by McCabe, Scotese, and others [Scotese et al., 1982; Wisniowiecki et al., 1983; McCabe et al., 1983, 1984] has convincingly demonstrated pervasive late Paleozoic

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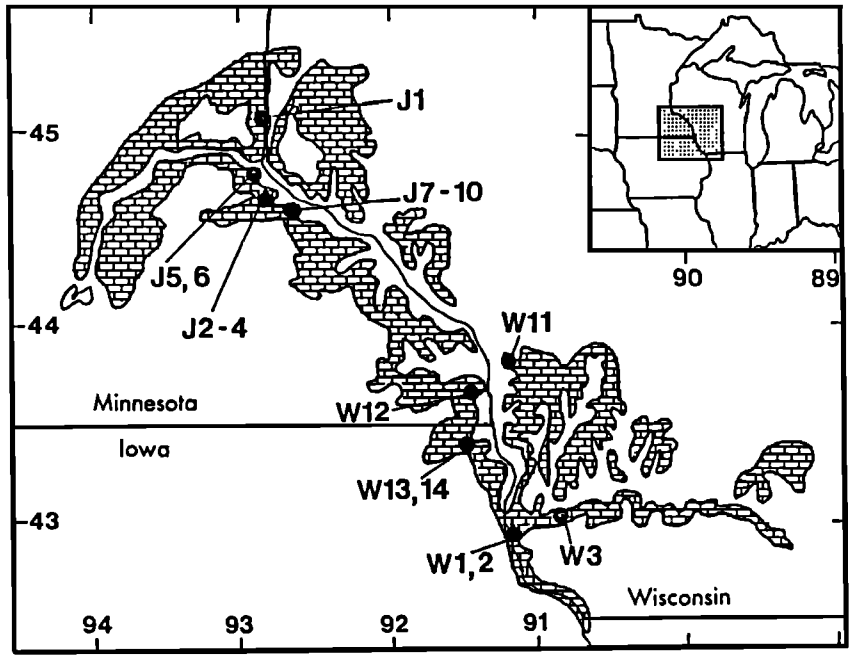


Fig. 1. Location of sampling sites for this study. Patterned area shows the outcrop belt of the Oneota dolomite. Inset map shows figure area.

remagnetization in lower and middle Paleozoic carbonates from widely separated localities on the craton. Despite these problems, however, we have investigated the paleomagnetism of the lower Ordovician Oneota dolomite, because its occurrence in the interior portion of the craton makes it likely to have escaped both tectonic rotation and the thermal/chemical remagnetizing effects associated with late Paleozoic orogenic activity on the margins of the craton.

Geology and Sampling

Lower Ordovician rocks occurring in the upper Mississippi River Valley are known as the Prairie du Chien group [Davis, 1966a]. The group is composed of three formations consisting dominantly of carbonates. In ascending order these are the Oneota dolomite, New Richmond sandstone, and Shakopee dolomite. The geology of the Prairie du Chien group has been summarized by Davis [1966b] and by Austin [1972].

The Oneota dolomite is a locally stromatolitic, fine- to medium-grained dolomite, incorporating basal sandstone and siltstone layers in the western half of the outcrop area [Austin, 1972]. The Oneota overlies the Upper Cambrian Jordan sandstone, with a gradational contact zone that seems to record continuous deposition across the Cambrian-Ordovician boundary [Kraft, 1956; Webers and Austin, 1972]. The Oneota is assigned a lower Canadian (Gasconadian) age by Twenhofel et al. [1954] and Heller [1956], representing a 5 million year interval from 500 to 495 Ma [Van Eysinga, 1975].

Above the basal clastic units the Oneota consists primarily of massive dolomites with a very minor fraction of terrigenous clastic material [Stauffer, 1950]. The contact between the Oneota dolomite and the overlying New Richmond sandstone is sharp. Fossil assemblages differ above and below the contact [Ulrich, 1924], and basal conglomerates [Webers and Austin, 1972] and truncation of beds [Andrews, 1955] are locally observed. Ludvigson and McAdams [1980] have presented structural evidence for a phase of regional epeirogenic tectonism prior to deposition of the New Richmond sandstone, possibly involving reactivation of high-angle basement structures. Twenhofel et al. [1954] and Heller [1956] assign a middle Canadian (Jeffersonian) age to the New Richmond sandstone and the Shakopee dolomite. Comparison with the Van Eysinga [1975] time scale indicates that approximately five million years elapsed between the end of Oneota deposition and the onset of New Richmond sedimentation.

The paleogeography of the region during the deposition of the Prairie du Chien carbonates has been described by Webers [1972] and by Davis [1966b]. The deposits were laid down in the broad, shallow Ancestral Forest City Basin. Isopach maps suggest that the basin opened seaward to the (present-day) south and west in Cambrian times and to the southeast during the Ordovician [Austin, 1972]. The Wisconsin Dome bounded the basin to the northeast and exposed the granites and sandstones which were the primary sources of detritus into the basin during Prairie du Chien sedimentation [Davis, 1966b].

Samples were cored in the field with a portable gasoline-powered drill during two field seasons. Doyle Watts collected 56 samples (7 sites) in northeastern Iowa and southeastern Wisconsin in 1976 (sites W1 - W3, W11 - W14, Figure 1). Laboratory analysis of these samples clearly showed the presence of a shallow, east-southeasterly or west-northwesterly characteristic magnetization. A further 79 cores (sites J1 - J10) were collected in 1982 in order to reduce the statistical uncertainty in the pole position and to attempt to obtain field evidence for the age of the characteristic remanence.

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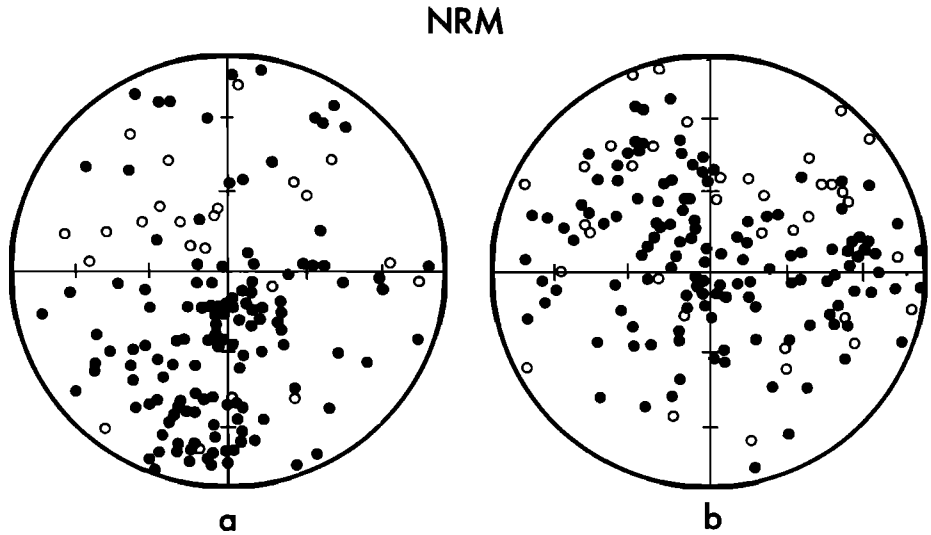


Fig. 2. Equal area projections of the directions of natural remanent magnetization in samples of the Oneota dolomite. Open symbols on upper hemisphere, solid on lower. (a) Sample coordinates, prior to correction for in situ core orientation. (b) In situ coordinates.

A potential field test for early remanence acquisition was provided by the occurrence of a basal breccia in the New Richmond sandstone in a small abandoned quarry near Miesville, Minnesota [Webers and Austin, 1972]. Angular clasts of Oneota dolomite with dimensions up to one meter are exposed in the quarry walls. Eight cores (site J3) were collected from six of these blocks for a conglomerate test. Due to limited accessibility it was not possible to sample more than six blocks. Sites J2 and J4 were located in undisturbed Oneota dolomite in the same quarry.

Paleomagnetic Analysis

The natural remanent magnetization (NRM) intensities were typically low, with a median value of 0.08 mA/m (8×10^{-8} emu/cm^3). NRM directions for the combined collection are strongly dispersed, but significantly nonrandom. Values of the uniform test statistic [Mardia, 1972] are 86.8 and 78.6, in sample and geographic coordinates, respectively. Randomness of directions can be rejected at the 1% level of significance in both cases. In sample coordinates (Figure 2a) the eigenvectors t1 and t3 corresponding to the minimum and maximum eigenvalues have declination/inclination D/I = 284/5 and D/I = 189/45 respectively, reflecting the visually apparent concentration of directions in a N-S vertical plane, with a maximum concentration in a southerly downward direction. A bipolar (unimodal) test statistic [Mardia, 1972] has a value of 1.62. In field coordinates (Figure 2b) t1 has D/I = 186/1, indicating concentration in an E-W vertical plane, and t3 has D/I = 280/77, as the directions are overwhelmingly downward. Bipolar and girdle test statistics are approximately 17 and 35 respectively, so the data do not conform to either distribution.

Methods

Samples were subjected to stepwise demagnetization by both thermal and alternating-field (AF) methods, using shielded Schonstedt instruments. All measurements were made on a two-axis ScT cryogenic magnetometer, with an estimated average noise level of 0.005 mA/m or less. Due to the low NRM intensities, several laborious noise reduction techniques had to be employed in order to maintain a satisfactory signal-to-noise ratio. Both random and systematic sources of error were of potential importance, including measurement error, short-period viscous behavior, chemical/mineralogical changes produced by heating, and acquisition of anhysteretic magnetizations during AF treatment. In order to reduce the effective noise level of the magnetometer, samples were routinely measured twice after each treatment and the two results averaged to obtain the magnetization vector. By inverting the sample between measurements and averaging appropriately, any contribution of the sample holder was canceled out. Similarly, anhysteretic components picked up during AF treatment were averaged out by performing each demagnetization step first along three mutually orthogonal axes, measuring, repeating the treatment along the three antiparallel axes, and measuring again [French, 1976; Johnson et al., 1984]. Thermal treatments were usually discontinued if laboratory-induced magnetization appeared significant, and care was taken to minimize the time and exposure to magnetic fields between treatment and measurement of the samples, in order to reduce the importance of short-period viscosity.

Results were plotted on orthogonal vector projections [Zijderveld, 1967] for analysis and identification of magnetic components. The criterion for isolation of components was collinearity of three or more consecutive points, as determined by visual inspection, with subjective allowance for noise. Directions were then determined by the principal components method of Kirschvink [1980]. In cases where the demagnetization trajectories were nonlinear or missed the origin, stereographic plots of the data were examined for coplanar segments, using similar

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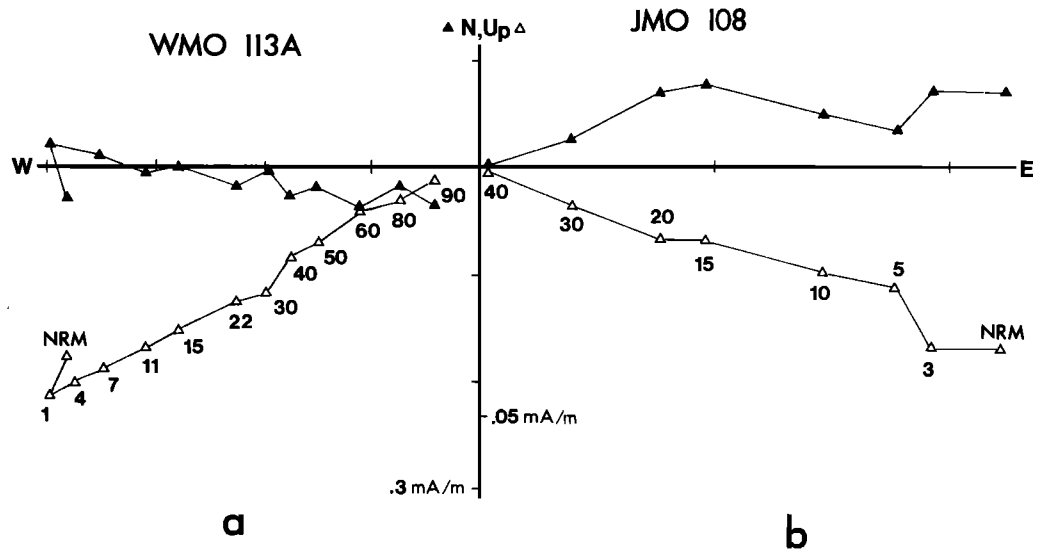


Fig. 3. Orthogonal vector projections [Zijderveld, 1967] of the characteristic shallow east-west component of magnetization, defined by AF demagnetization. Solid symbols represent successive positions of the end point of the magnetization vector during stepwise demagnetization projected onto the horizontal (N,E) plane. Open symbols are projections onto the vertical (E,Up) plane, and are labeled with peak AF values in milliteslas. Sample WMO 113A is from site W12, JMO 108 from site J10.

criteria. The direction of the normal to the plane was determined by the least squares method of Halls [1976, 1978].

Results

Characteristic directions. A well-separated characteristic shallow E-SE/W-NW component, clearly identifiable on orthogonal vector projections, was present in 50 of the 149 analyzed specimens (Figures 3 and 4). During AF demagnetization the characteristic component was usually completely removed in fields of 50 to 80 mT, with a median destructive field (MDF) most often between 10 and 15 mT, but commonly as high as 30 mT (Figure 3). A small higher-coercivity

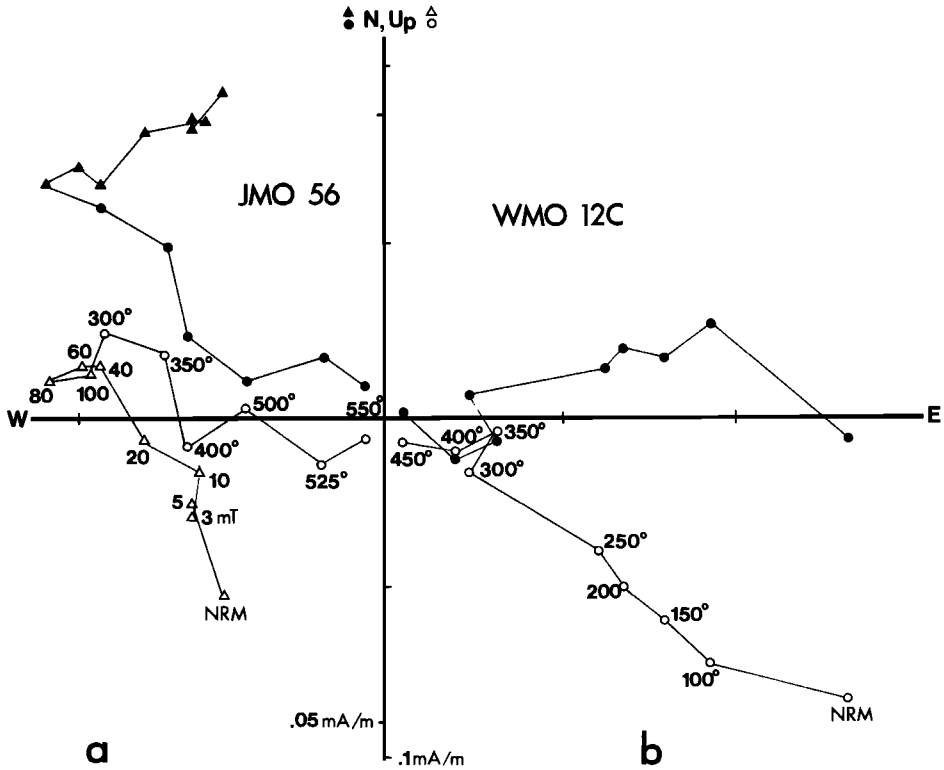


Fig. 4. Orthogonal projections of the characteristic magnetization defined by thermal demagnetization. Triangles show magnetization during AF treatment, labeled with peak fields in milliteslas. Circles show thermal demagnetization, temperatures in degrees Celsius. WMO 12C is from site W2, JMO 56 from site J5.

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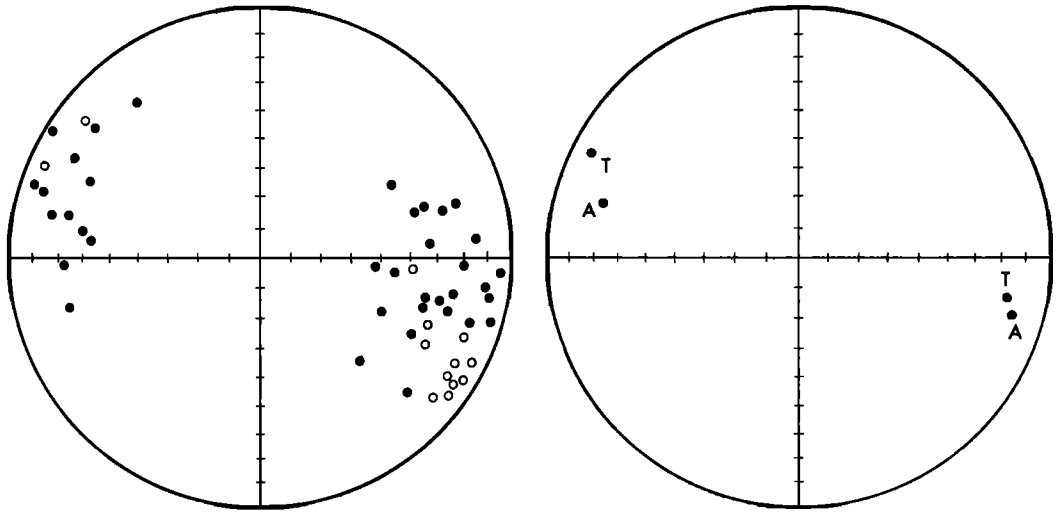


Fig. 5. (a) Equal area plot of sample characteristic directions isolated on vector plots as in Figures 3 and 4. (b) Mean directions of the normal- and reversed-polarity groups, as determined by thermal (T) and AF (A) demagnetization. Both the normal and reversed polarity means have positive inclinations of 14°, and thermal and AF methods yield virtually identical results.

component was present in about 10% of the analyzed specimens, but its direction could usually not be determined (Figure 3a). Only samples from sites J5 and J6 had a measurable magnetization remaining after treatment in the maximum available AF of 100 mT, and subsequent thermal demagnetization of these samples isolated the characteristic component in two out of three cases (Figure 4a).

During thermal demagnetization, the characteristic component was almost always the highest unblocking-temperature (T_{ub}) component if more than one were present, with T_{ub}'s smoothly distributed up to a typical maximum of 350° to 450° C, with no measurable magnetization remaining in any sample after heating to 550° C (Figure 4). Between 300° and 400° C most samples underwent some chemical/mineralogical changes, taking on a pronounced reddish color. This probably represents production of hematite, possibly from goethite or maghemite [Lowrie and Heller, 1982], and was commonly accompanied by acquisition of small spurious components of magnetization (e.g., sample WMO 12C, Figure 4). Data points which were seemingly affected by such chemical/mineralogical changes were disregarded in the determination of a sample's characteristic direction of magnetization.

For approximately 10 to 20% of the samples, particularly those from sites J2 - J4, a large component of magnetization with a direction close to that of the present-day field (PDF) was removed by 150° C. AF demagnetization was less effective in removing this component. The low T_{ub} and high coercivity, as well as the direction, suggest that this is a relatively recent magnetization carried by goethite. Subsequent behavior of these samples during progressive cleaning was noisy, and the characteristic component could not be isolated in any of them. Most samples from the remaining sites had small or moderate randomly oriented soft components, which were removed by 150° C or 10 mT.

The general behavior during demagnetization, with moderate T_{ub}'s and coercivities, suggests that the characteristic remanence is carried dominantly by magnetite. The characteristic directions fall in two nearly antipodal groups (Figure 5a); two-thirds of them are east-southeasterly and are interpreted as reversed polarity. Both the normal- and reverse-polarity groups have predominantly positive inclinations, which may be due to contamination by an incompletely separated downward component. This will be considered further in a later section.

TABLE 1. Mean of Sample Characteristic Directions

Treatment	Polarity	Mean Decl/Incl	N	k
AF	normal	287/20	9	17.7
Thermal	normal	296/08	6	15.5
AF	reverse	106/13	15	7.8
Thermal	reverse	103/16	20	7.5
Both	normal	290/14	15	13.8
Both	reverse	104/14	35	7.8
overall mean		106/05	50	7.4
		$\alpha 95 = 8.0°$		

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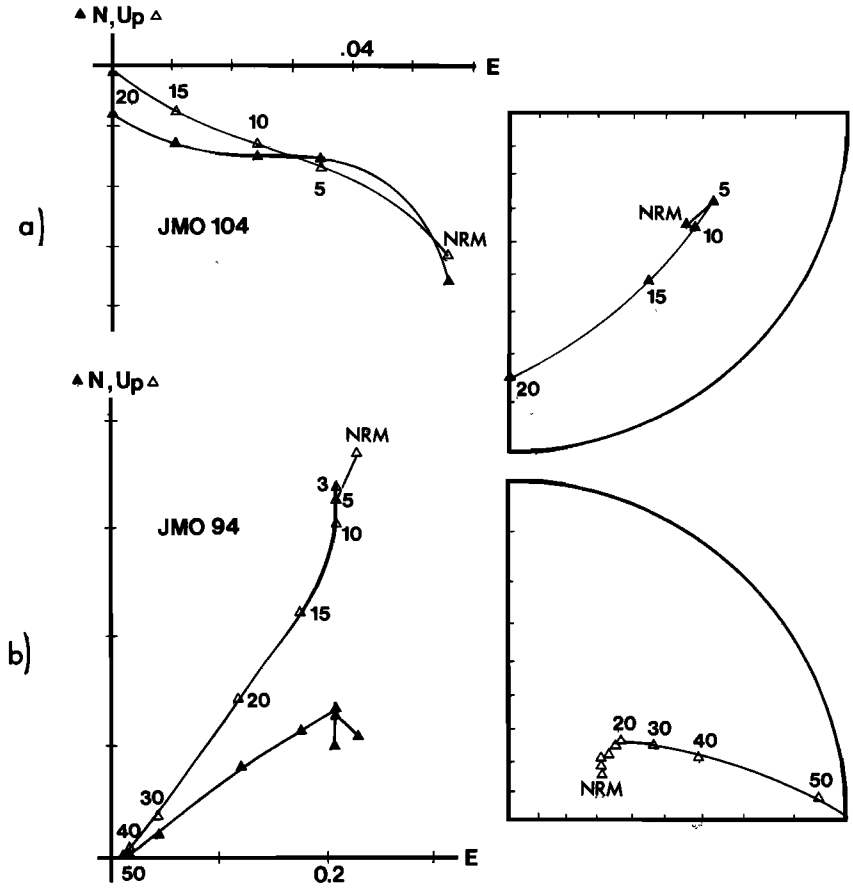


Fig. 6. Examples of overlapping multicomponent magnetizations, identified by (a) nonlinear demagnetization trajectory and (b) linear trajectory bypassing the origin; both of which clearly define great-circle trajectories, as shown on the right. Axes labeled in units of milliamperes per meter, demagnetization data labeled with peak AF in milliteslas.

Thermal and AF cleaning produced virtually identical results (Figure 5b). These results are summarized in Table 1.

Remagnetization circles. For approximately 65% of the samples in this study, the characteristic component of magnetization could not be isolated on orthogonal projections. These vector diagrams typically showed either a nonlinear trajectory (Figure 6a) or a linear trajectory which missed the origin (Figure 6b), indicating in the first case simultaneous removal of two or more components with overlapping stability spectra, and in the second case suggesting the presence of a stable component whose intensity is too low for resolution. In stereographic projection the trajectory of the magnetization vector usually described a great-circle path which passed close to the shallow E-SE/W-NW characteristic direction (Figure 6). Least squares analysis [Halls, 1976] was used to compute the best fit great-circle path for each sample. Due to the low magnetic intensities these were rather noisy, and Q values were typically about 10^{-4}, indicating a moderately good fit. For about 35 samples, especially those from site W11, and most often those which had been treated thermally, no linearity or planarity was identifiable in the demagnetization data, and these will not be considered further.

The distribution of the poles to the best fit great circles is shown in Figure 7 and clearly defines a roughly north-south vertical plane. The least squares pole to this in turn provides the best estimate of the characteristic component [Halls, 1976]. The direction determined for 48

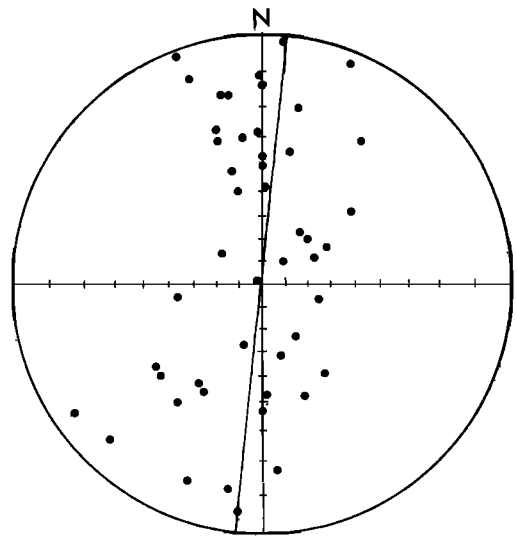


Fig. 7. Equal angle plot of great-circle poles, each point representing one specimen. Curve shows least squares best fit plane.

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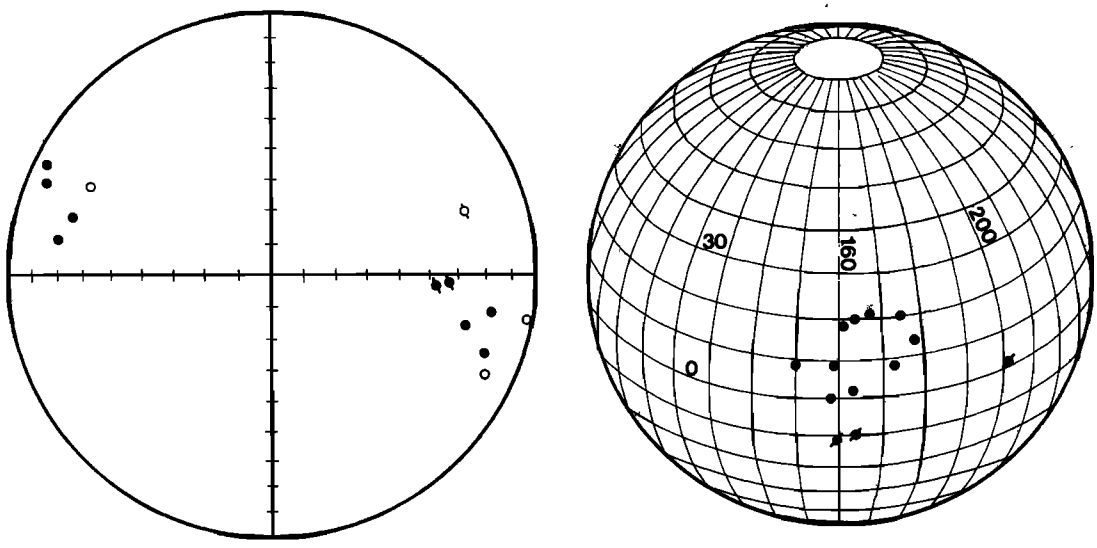


Fig. 8. Site means: (a) characteristic directions; (b) VGP's. Sites W14, J6, and J8,9 indicated by slashes.

samples has D/I = 95/-1. The ellipse of 95% confidence, computed by the method of Bailey and Halls [1984], has radii of 12 and 16 degrees, overlapping the 95% confidence circle about the mean of the characteristic directions (Table 1).

Site means (Figure 8, Table 2) were computed for sites where the characteristic component could be isolated in at least four samples, or where at least four sample great circles converged near the characteristic direction. These criteria were met by 13 of the 17 sites. Means were computed using the method of Bailey and Halls [1984] to incorporate both great-circle and vectorial data. Site W11 contained only unstable magnetizations, and great circles for site J1 converged at the PDF direction, so these sites have been discarded. The great-circle poles for samples from site J8 fell in a tight northerly, horizontal group, indicating the presence of two coherent components but preventing determination of the direction of either one. The data for this site have been combined whith those for site J9. The mean directions and site virtual geomagnetic poles (VGP's) cluster fairly well, with three outliers: sites J6 and W14 have anomalously steep directions, while the direction for the combined

TABLE 2. Site Means

Site	Mean D/I	VGP (°N/°E)	k	ER1	ER2	Nt(Ns)/Na	N/R
W1	94/17	-3/169	9	28		(6)/10	2/4
W2	105/25	2/159	8	17		(12)/15	0/12
W3	106/-13	19/162	11	21		(6)/9	0/6
W12	282/22	17/179	9	33		(4)/5	4/0
W13	284/13	15/174	16	21		(5)/7	2/3
W14*	95/37*	-11/160*	26	8		(11)/11	0/11
J2	296/-7	16/156	18	26	37	4(1)/8	1/0
J4	104/27	0/158	7	37	60	7/11	
J5	292/16	21/167	14	15	20	8(3)/11	3/0
J6*	92/31*	-10/164*	3	...	...	4/7	
J7	282/15	14/174	9	48	53	4(2)/8	2/0
J8,9*	71/-26*	-3/201*	68	6	11	14/15	
J10	99/13	2/166	46	10	11	7(2)/9	0/2

mean 105.4/+2.0 (k = 17.6, α95 = 11.9°, N = 10)
mean VGP 10.4° N, 166.4° E (K = 48.0, A95 = 7.0°, N = 10)

ER1, ER2: radii of the ellipse of 95% confidence; for sites with no great circle data, ER1 equals Fisher's α95. Nt: number of samples yielding great circles. Ns: number of samples yielding stable endpoint direction. Na: number of samples analyzed. N/R: number of normal/reverse polarity characteristic directions.

* Not used in computing mean.

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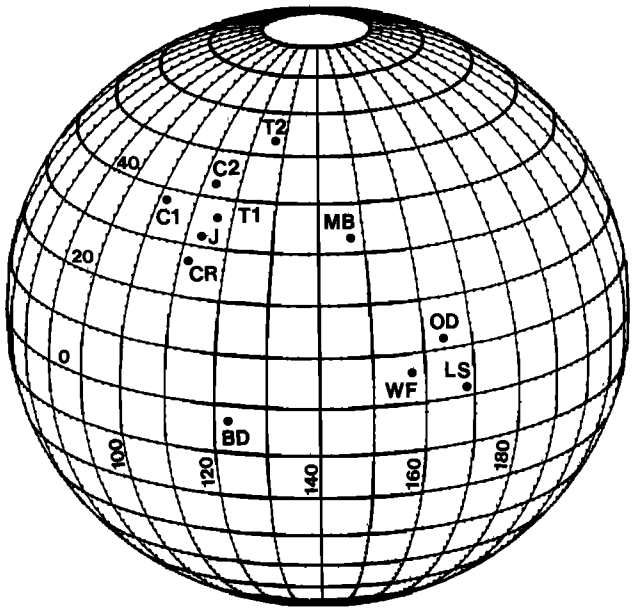


Fig. 9. Selected paleomagnetic pole positions from Ordovician and late Cambrian rocks: J, Juniata formation, upper Ordovician [Van der Voo and French, 1977]; MB, Moccasin-Bays, middle Ordovician [Watts and Van der Voo, 1979]; CR, Chapman Ridge, middle Ordovician [Watts and Van der Voo, 1979]; T1, Trenton limestone, middle Ordovician [McElhinny and Opdyke, 1973]; T2, Trenton limestone [McCabe et al., 1984]; OD, Oneota dolomite [this study]; C1, Colorado intrusives, 497±16 Ma [Larson et al., 1985]; C2, Colorado intrusives, 495±10 Ma [Lynnes and Van der Voo, 1984]; BD, Buckingham (Quebec) dikes, 497±24 Ma [Dankers and Lapointe, 1981]; LS, Lamotte sandstone, upper Cambrian [Al-Khafaji and Vincenz, 1971], recalculated by Van der Voo et al. [1976]; WF, Wilberns formation, upper Cambrian [Van der Voo et al., 1976].

sites J8 and J9 is much more northerly than the others. These have not been used in computing the formation mean. The mean of the site means is D/I = 105/+2 (k = 17.6, $\alpha 95$ = 11.9°, for N = 10 sites), and the corresponding paleomagnetic pole falls at 10.0° N, 166.4° E, dp = 5.7°, dm = 11.4° (Figure 9). The mean of the site VGP's is 10.4° N, 166.4° E, K = 48.0, A95 = 7.0°.

Breccia test. As was the case for many of the in-situ samples, the response of breccia block samples (site J3) to stepwise cleaning was usually a linear decay of the magnetization vector toward a point removed from the origin (Figure 10), suggesting the presence of a relatively high-stability, low-intensity component. The direction of this stable component can be recovered by great-circle analysis only if its dispersion is smaller than that of the soft component. Of the eight samples, two were subjected to thermal treatments and yielded incoherent results. The remaining six, however, produced identifiable great circles (Figure 11, Table 3). The poles to these great circles (Figure 11) lie in a plane normal to the large northerly, steep component identifiable in Figure 10. This large soft component is therefore less dispersed than the high-stability component. Further, the poles are widely spread out along the great circle normal to the soft component. A statistical test for a symmetric girdle distribution [Mardia, 1972] applied to the great-circle poles yields a test statistic of 2.9, strongly suggesting a distribution with rotational symmetry. Thus the directions of the stable component are strongly dispersed, and a necessary condition for a positive conglomerate test [Graham, 1949] is met. However, in order to obtain a conclusive test it is also necessary to demonstrate that the directions of the stable component in multiple samples from the same clast are coherent. In the context of great-circle analysis, this means that the poles to great circles determined for two or more samples from

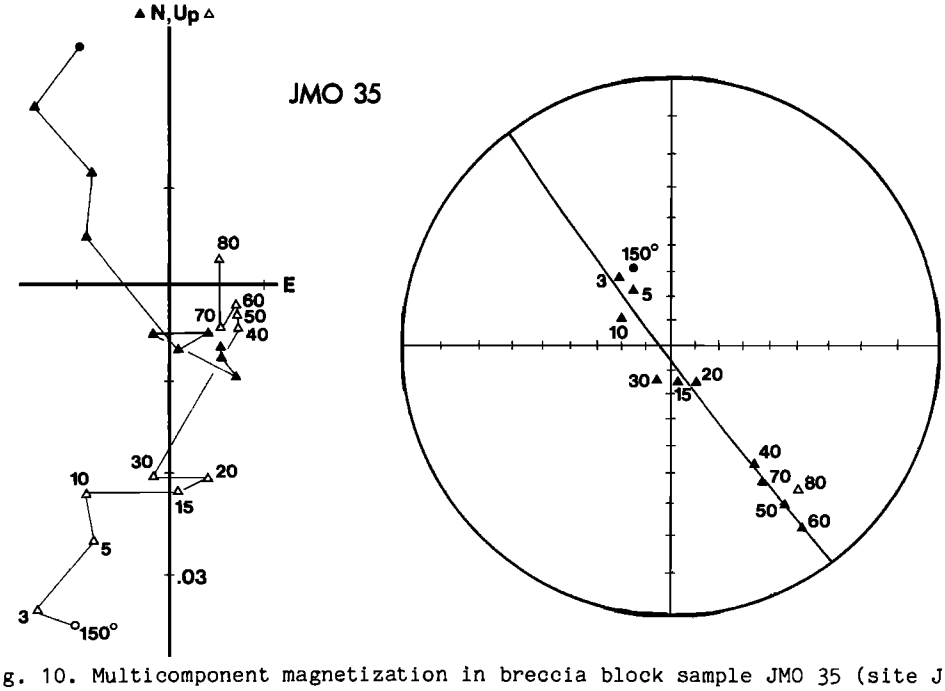


Fig. 10. Multicomponent magnetization in breccia block sample JMO 35 (site J3).

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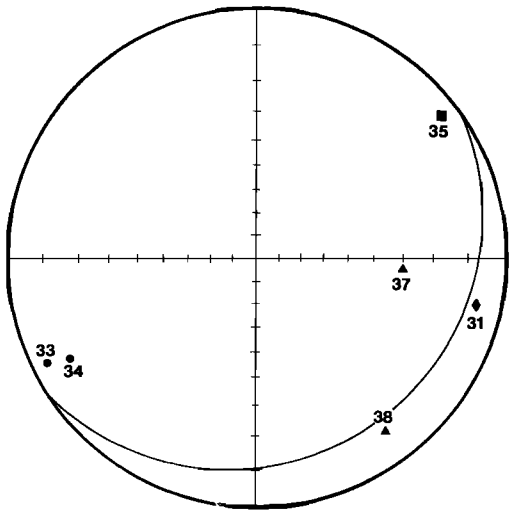


Fig. 11. Distribution of great-circle poles for samples from breccia. Different symbols indicate different breccia blocks. Solid curve shows the least squares best fit plane. The direction of the common component is northerly and steeply downward.

the same block should lie together, 90° from both the characteristic and secondary components. While the distribution of the great-circle poles in Figure 11 is crudely consistent with this requirement, it is clear that the resolution of directions is not sufficient to enable us to claim a conclusive conglomerate test.

The test for remanence acquisition prior to brecciation would be negative if the great circles converged near the shallow E-SE/W-NW characteristic direction. This is clearly not the case. The test of Mardia [1972] for coplanarity with the specified normal axis yields a test statistic of 7.16. The breccia test is therefore nonnegative at the 95% confidence level. Our interpretation of the results from the breccia blocks is that they favor acquisition of the stable remanence in the Oneota dolomite prior to deposition of the New Richmond sandstone (approximately 490 Ma), but that the resolution of the direction of the stable component in the breccia samples is not good enough to make the test conclusive.

Magnetic Mineralogy

When samples were given an isothermal remanent magnetization (IRM) with an electromagnet in a stepwise manner up to the maximum attainable field of 0.9 T and then thermally demagnetized, three distinct patterns of behavior were observed. The first of these occurred in samples from sites J9 and J10 and is characterized by a very steep initial slope which flattens abruptly at 0.05 T (Figure 12a). For sample JMO 108, 96% of the total IRM was acquired by 0.05 T, and the IRM curve then remains flat to 0.9 T. The very low coercivity phase which dominates the remanence of these samples is most likely magnetite, with predominantly weak anisotropy, either multidomain (MD) or nearly equant single-domain (SD) grains [Evans and McElhinny, 1969]. The failure of JMO 108 to acquire any further IRM in fields up to 0.9 T rules out a significant content of hematite, although goethite may remain unactivated in fields up to 1 T [Lowrie and Heller, 1982] and may therefore be present in these samples.

The second type of behavior was exhibited by samples from site J3 (Figure 12b). IRM acquisition in samples which had previously been thermally demagnetized to 350° or 400° C indicates much less high-coercivity material than in those which had not. Over one half of the total IRM was removed by heating to 100° C for previously unheated JMO 34, and the magnetization was completely unblocked by 500° C. For sample JMO 31, which had previously been subjected to thermal cleaning, the reduction in IRM intensity caused by heating to 100° C was small, and unblocking temperatures extended up to at least 600° C. The high coercivities and low T_{ub}'s in the unheated samples clearly indicate the presence of goethite in site J3, and it is apparent that heating to 350° C causes the goethite to invert to hematite [Lowrie and Heller, 1982].

The third type of behavior was observed in samples from sites J2, J5, W2, and W14. IRM acquisition curves show a slightly higher proportion of high-coercivity material in heated samples than in unheated samples (Figure 12c). The difference in intensity between heated and unheated samples is very small after thermal treatment at 100° C, becoming larger by 300° C, with higher intensities in the previously heated samples. This suggests that goethite is

TABLE 3. Great Circle Poles from Blocks in Conglomerate

Block	Sample	Pole Decl/Incl	N	Q
A	31	112/6	6	1.8×10^{-3}
B	32	x		
C	33	244/4	6	1.7×10^{-4}
	34	242/8	8	3.8×10^{-3}
D	35	53/4	13	3.2×10^{-6}
E	36	x		
F	37	94/30	6	9.1×10^{-4}
	38	143/9	13	5.3×10^{-4}

N: number of coplanar data points; x: no linearity or planarity in demagnetization data.

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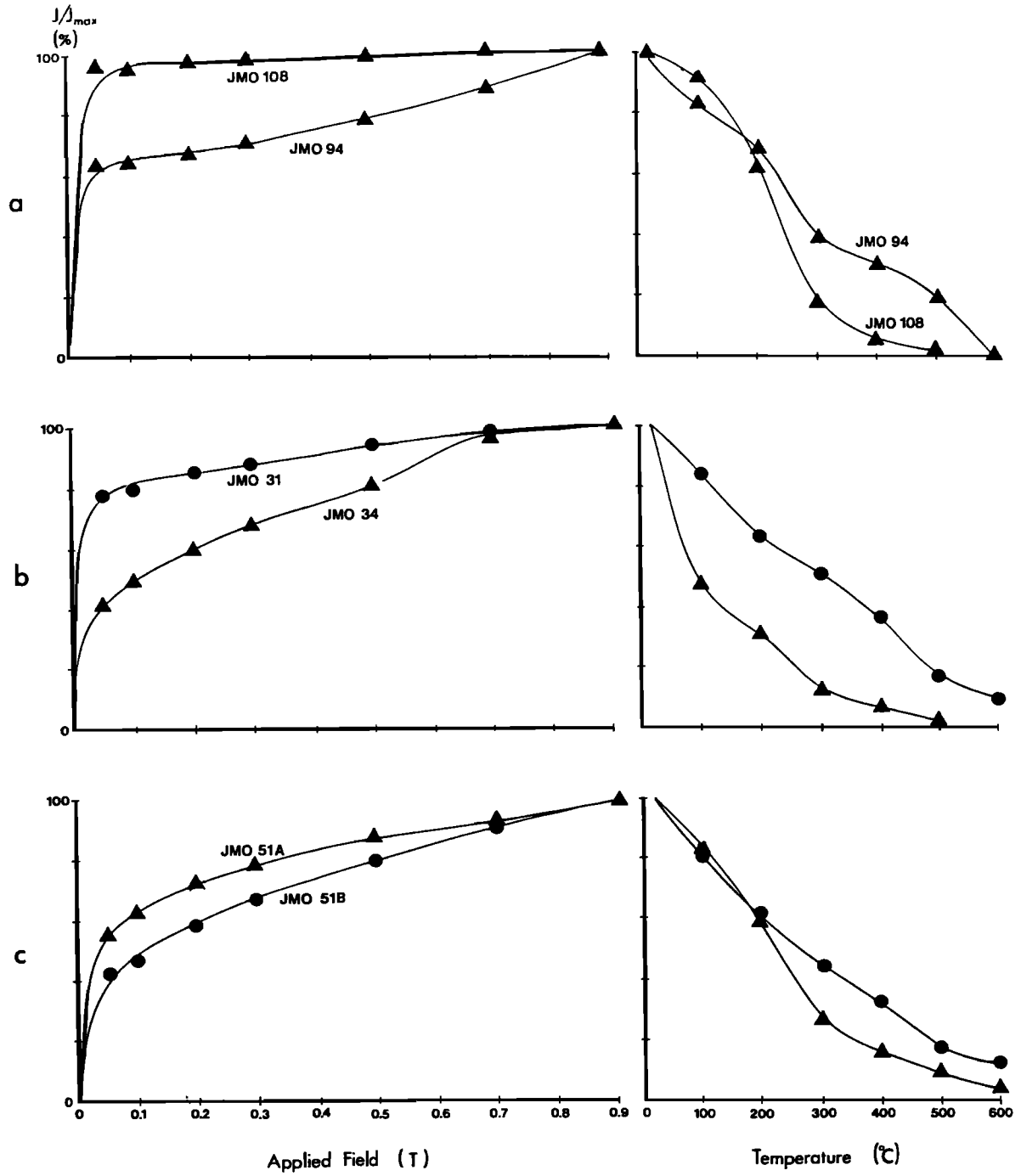


Fig. 12. Acquisition and thermal demagnetization of IRM for samples previously subjected to thermal demagnetization to 350° or 400° C (circles) or to AF cleaning (triangles). Samples from (a) sites J9, J10; (b) site J3; (c) site J5.

magnetically unimportant in these samples (although as previously mentioned it may have simply remained unmagnetized in the maximum available field), that hematite is initially present, and that additional hematite may be produced by heating to 350° C.

Discussion

While the two groups of characteristic directions in Figure 5 are nearly antipodal, there is a downward bias in the directions, as both the normal- and reverse-polarity groups have predominantly positive inclinations. When all of the characteristic directions are included, the mean for each group has a positive inclination of 14° (Table 1). The deviation from perfect antiparallelism is thus 28°.

Although it is conceivable that this downward bias could be explained in terms of long-term nondipole components in the earth's magnetic field [Coupland and Van der Voo, 1980], asymmetric field

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reversals [Nevanlinna and Pesonen, 1983], or very rapid APW during deposition of the Oneota dolomite, we consider it more likely that the downward bias is due to incomplete separation of magnetic components in the samples.

We have found evidence in other carbonate rocks, including the Shakopee dolomite [Jackson and Van der Voo, 1985], that spurious components of magnetization can be acquired during drilling and sawing of the specimens, and that these can produce a systematic bias in directions, at least as large as that observed in the Oneota samples. The unblocking-temperature and coercivity spectra of the drilling-induced remanent magnetization (DIRM) overlapped so strongly with those of a laboratory-induced anhysteretic remanent magnetization (ARM), that stepwise demagnetization produced apparent univectorial decay, deviating from the "primary" ARM direction by as much as 50° in the Shakopee samples. The angle of deviation correlated strongly with the ratio of initial to anhysteretic susceptibility χ/χ_{ARM}, which has been found by Banerjee et al. [1981] and King et al. [1982] to be sensitive to magnetite grain size, suggesting that DIRM acquisition depends on the presence of large (>10 μm) magnetite grains.

Because of the presence of significant amounts of hematite and, for some sites, goethite in the Oneota samples, the χ/χ_{ARM} method is inapplicable. It is logical, however, to suppose that the distribution of magnetite grain sizes in the Oneota dolomite may resemble that in the overlying Shakopee dolomite in having a significant fraction of large grains. This is supported by the presence of relatively abundant low-coercivity material (Fig 12).

In our study of DIRM in the Shakopee dolomite, we found that the DIRM was acquired in a direction parallel to the ambient field during cutting. The field during drilling was close to vertical (downward), while during sawing it was southerly and horizontal, in sample coordinates. On inspection of Figure 2a it is evident that the NRM directions tend to cluster near these cutting-field directions, suggesting a fairly strong DIRM overprint. After correction for the in-situ orientation of the cores, the DIRM components are directed downward in all cases, with moderate inclinations and dispersed declinations. The dispersion of the DIRM component and the overlapping stability spectra of the DIRM and characteristic components account for the success of great-circle methods in isolating the characteristic component.

Because we were unable to remove the downward bias with the cleaning procedures used, it was necessary to remove it mathematically. The process of averaging the dual-polarity results is the simplest way to accomplish this, although it may not be completely effective, because the number of reversed directions is more than twice the number of normal directions. The symmetry in the distribution of characteristic directions, with both the normal- and reverse-polarity groups having positive inclinations of 14°, suggests that the true inclination of the characteristic component may be 0°, with an equal angular bias in both sets. However, it is entirely possible that the bias in one set is larger than that in the other. With no independent means available to evaluate the relative magnitudes of uncleaned secondary components in the normal- and reverse-polarity sets, we have taken the mean of the site-mean directions, excluding the previously discussed outliers, to be the best estimate of the characteristic direction; this yields a mean inclination of +2° (Table 2).

Summary and Conclusions

The paleomagnetic pole position we have determined for the Oneota dolomite lies a considerable distance from the late Paleozoic portion of the APW path, and from any poles from younger rocks. There is thus no reason to suspect that the magnetization has been reset. Little is known about the effects of dolomitization on the magnetization of limestones, but because it is a pervasive process, involving recrystallization and perhaps substantial volume changes in the bulk rock [Blatt et al., 1972], it may strongly affect the magnetization. We have argued that the conglomerate test for the Oneota dolomite, while not conclusive, favors acquisition of remanence prior to deposition of the New Richmond sandstone (approximately 490 Ma). Whether the remanence is primary, or whether the Oneota magnetization was reset by dolomitization and/or epeirogenic tectonism during the 5-million-year interval prior to New Richmond sedimentation is a matter for speculation, but in either case we feel reasonably confident that the remanence was acquired very early.

The quiescent Phanerozoic history of the region makes it very unlikely that subsequent tectonic rotations have affected the direction of magnetization in the Oneota dolomite, and the consistency of magnetic directions over the large sampling area conclusively rules out any such consideration. The pole position for the Oneota dolomite is thus free of the two major uncertainties which plague other lower Paleozoic results, rotation and remagnetization.

Paleomagnetic data published to date for lower Paleozoic North America indicate a net counterclockwise rotation of the craton by about 65 degrees over the Cambrian-Ordovician interval. Larson et al. [1985] have argued that virtually all of this rotation was complete by late Cambrian time and that APW for the rest of the early Paleozoic is insignificant. Our results, in contrast, indicate that about 45 degrees of counterclockwise rotation occurred during the Ordovician. Further careful work on Ordovician rocks, with particular concern for the timing of remanence acquisition, is essential in order to resolve the early Paleozoic rotation history of North America.

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